

PolyRigorous — ThermoExplorer notebook printout

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PolyRigorous — Thermodynamics Explorer

Interactive exploration of the Flory-Huggins and Sanchez-Lacombe polymer solution thermodynamics implemented in the `PolyRigorous` package (`../src`). This notebook is a thin UI on top of that package — all the physics lives in the package, not here, so it stays testable independent of Pluto.

1. Choose a polymer / solvent pair

Selected pair: **Polystyrene** in **Toluene**, at 25 °C (298.2 K).

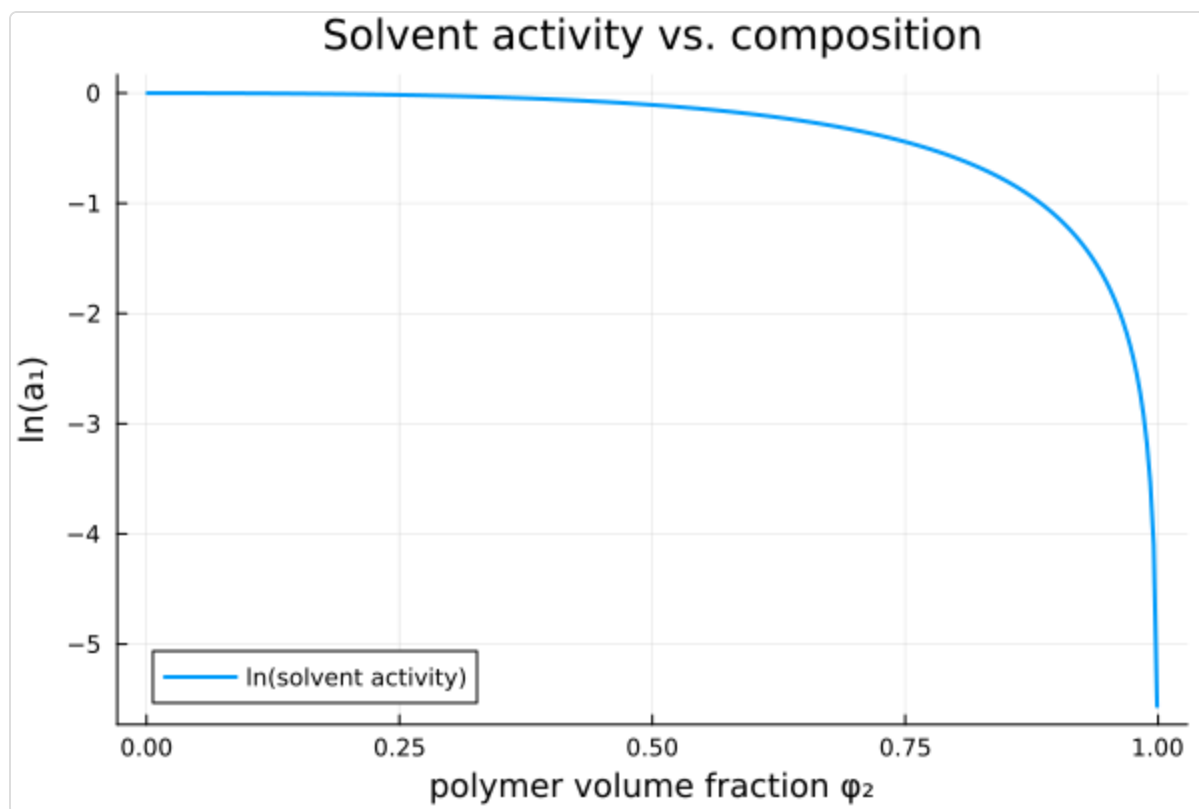
Flory-Huggins size ratio $N = V_m(\text{polymer})/V_m(\text{solvent}) = \mathbf{888.7}$

2. Flory-Huggins interaction parameter χ

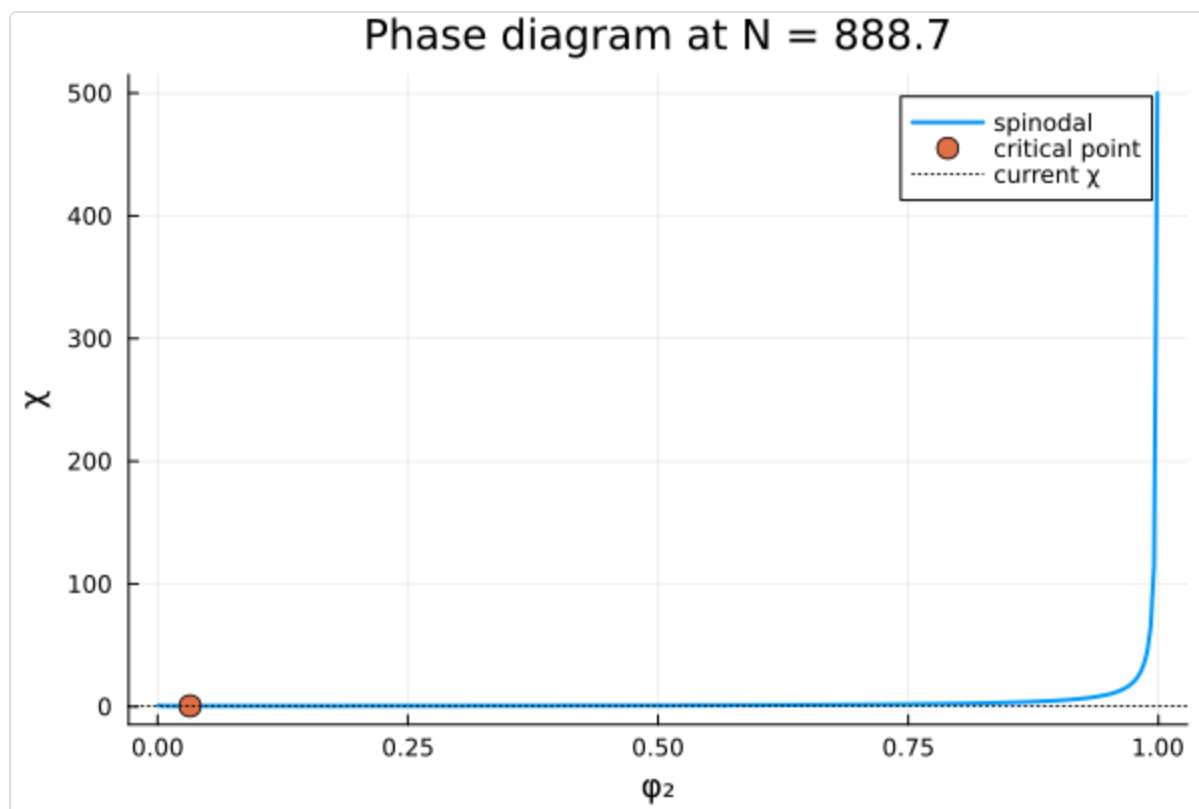
χ (from solubility parameters, entropic correction $\chi_s = 0.34$) = **0.347**

Critical point for this N : $\phi_{2,c} = 0.0325$, $\chi_c = 0.534$

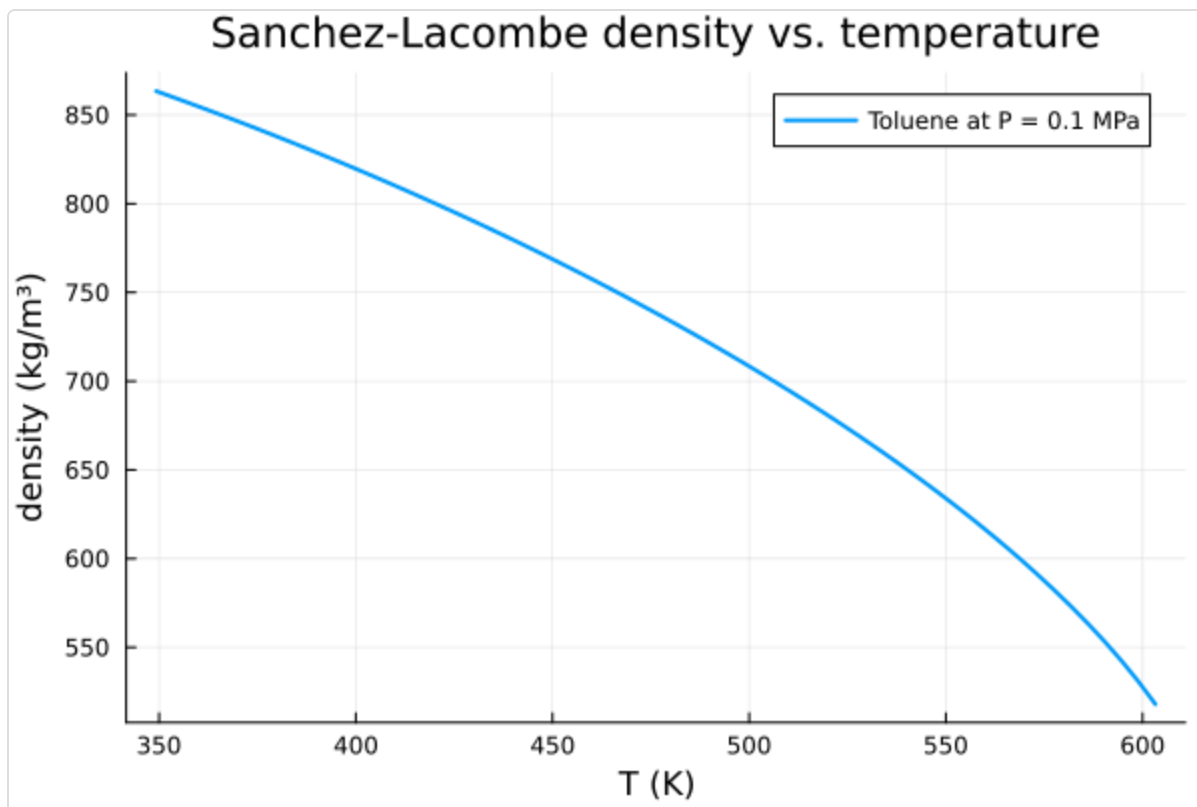
✓ $\chi < \chi_c$ at this temperature: the model predicts complete miscibility at all compositions.



3. Phase diagram (χ vs. ϕ_2)



4. Sanchez-Lacombe pure-component PVT



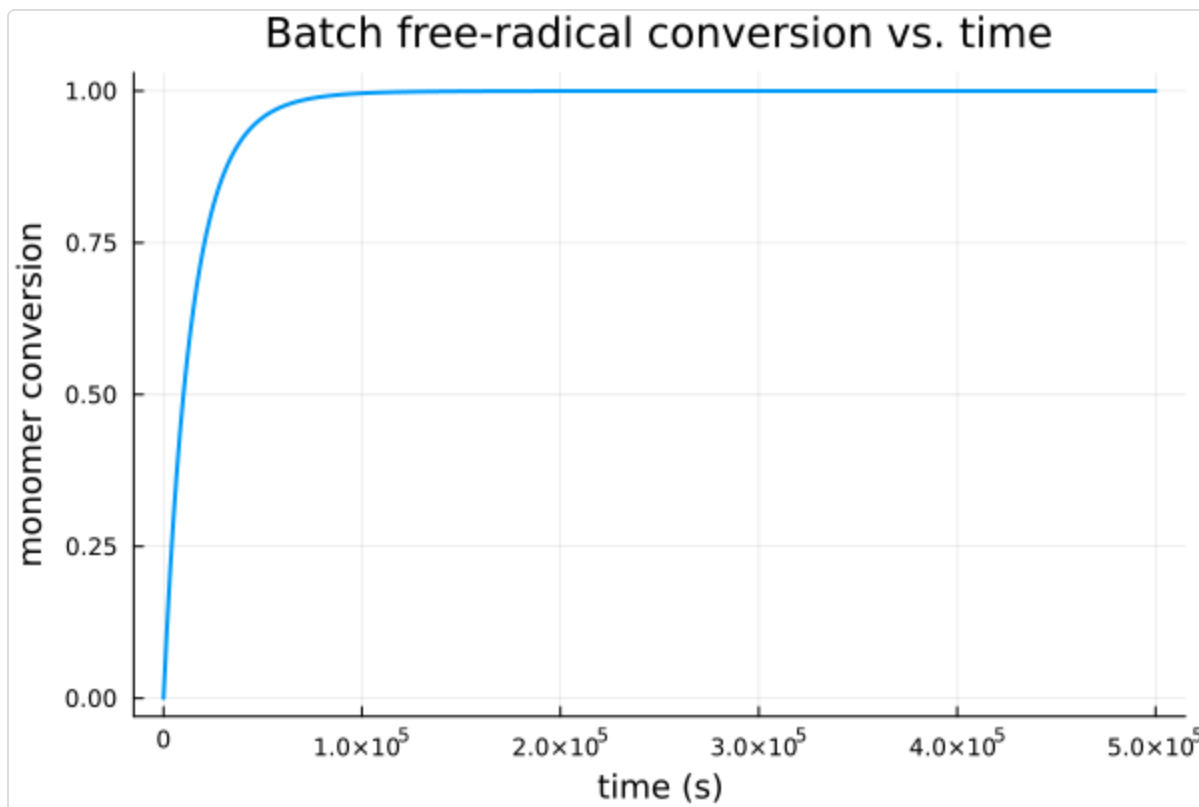
5. Free-radical polymerization kinetics

Batch, isothermal free-radical polymerization under the standard quasi-steady-state approximation (QSSA), from `kinetics_free_radical.jl`. Propagation (k_p) and termination (k_t) rate constants are held fixed at typical styrene-like values; vary the initiator decomposition rate k_d , initial initiator concentration $[I]_0$, and the disproportionation fraction δ of termination events.

$k_d = 1.0\text{e-}5 \text{ 1/s}$, $[I]_0 = 0.01 \text{ mol/L}$, $\delta = 0.2$

Kinetic chain length $\nu = 3535.5$; number-average degree of polymerization $X_n = 5892.6$

(Instantaneous PDI is exactly 1.5 for pure combination ($\delta=0$) and 2.0 for pure disproportionation ($\delta=1$); this package does not yet provide a closed-form PDI for the intermediate mixed case.)

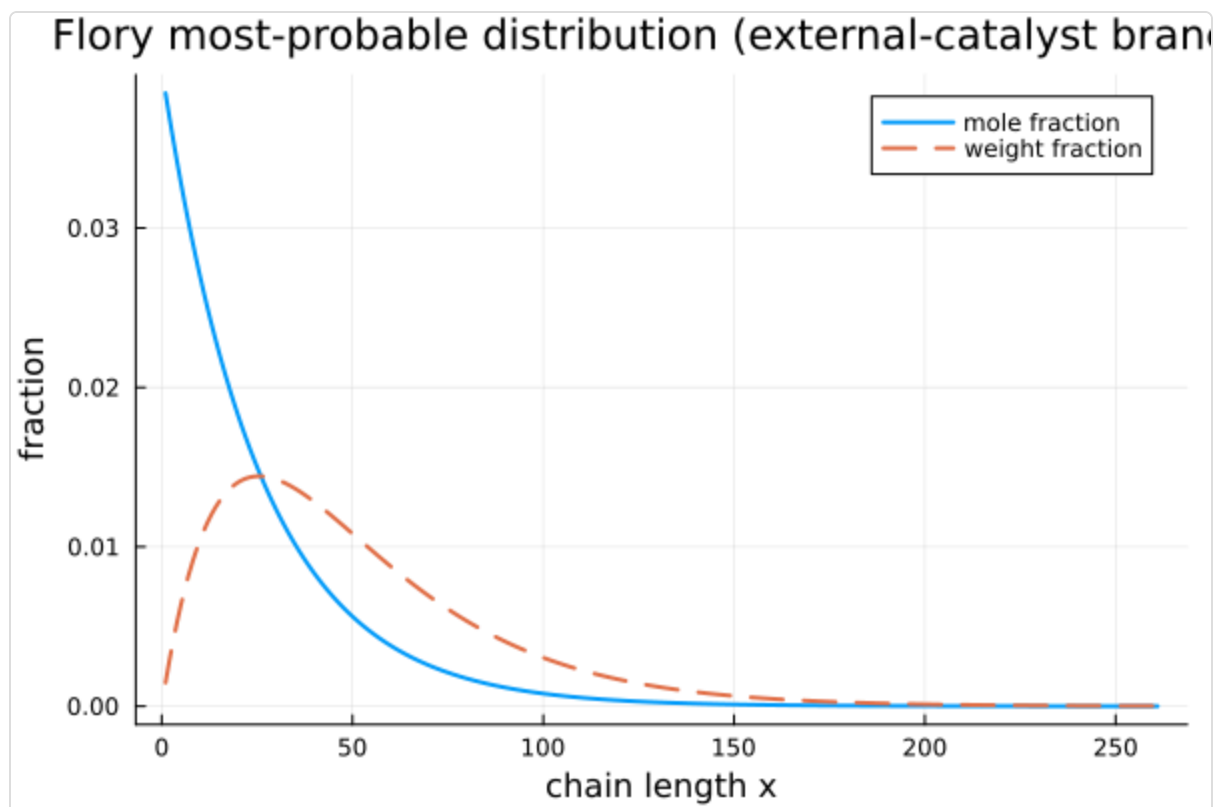
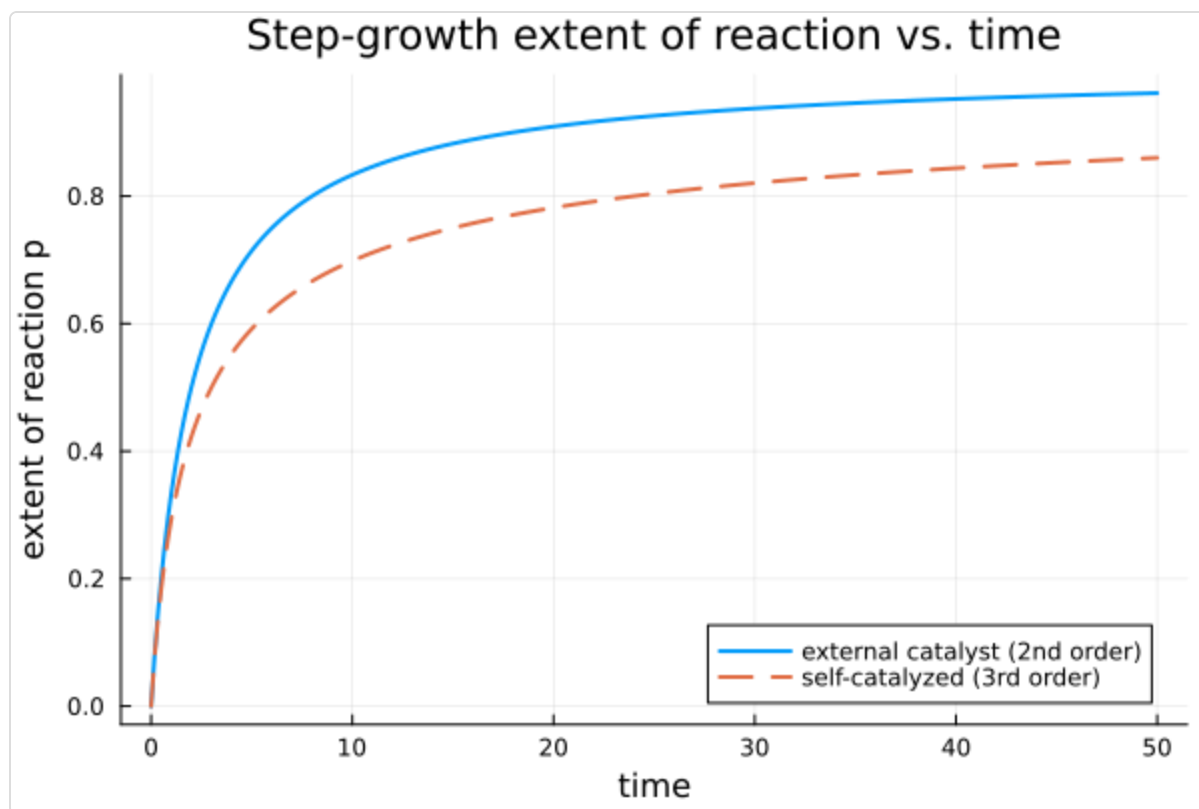


6. Step-growth polymerization & the Flory distribution

Extent of reaction $p(t)$ for the two classic step-growth kinetic cases (externally catalyzed = second order overall; self-catalyzed = third order overall), from `kinetics_step_growth.jl`, plus the resulting Flory "most probable" molecular weight distribution at the extent of reaction reached by time t_{max} .

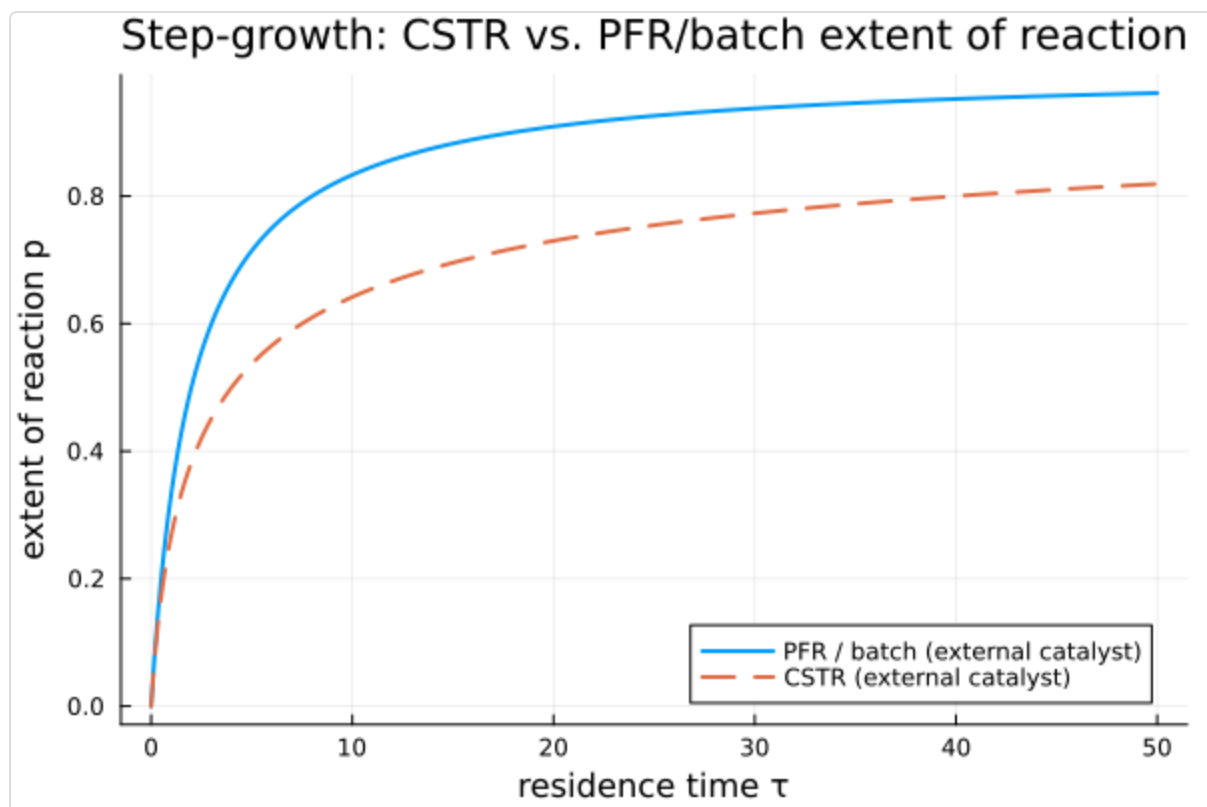
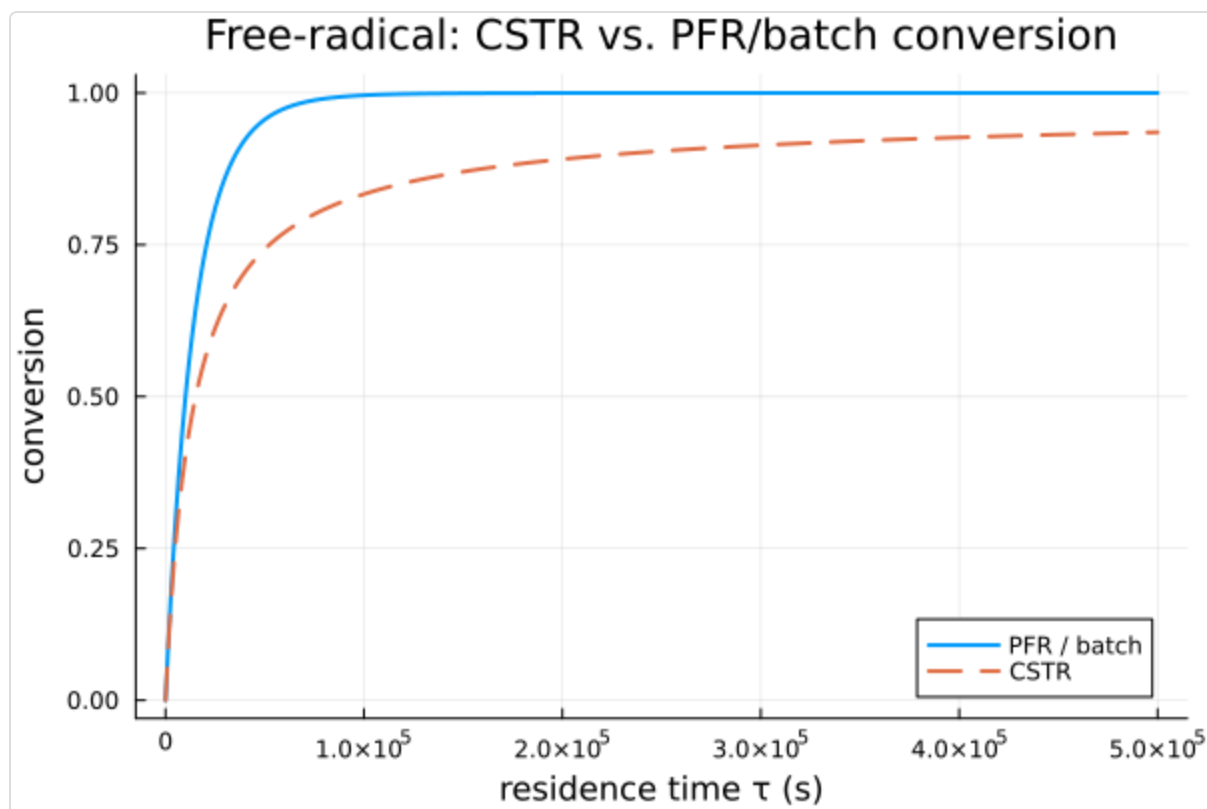
At $t = 50$: p (external catalyst) = 0.9615, $X_n = 26.0$; p (self-catalyzed) = 0.86, $X_n = 7.1$.

$\text{PDI} = 1 + p \rightarrow 1.962$ (external catalyst case), approaching 2 as $p \rightarrow 1$.



7. Reactor models: batch/PFR vs. CSTR

An ideal PFR is kinetically equivalent to a batch reactor run for a time equal to its residence time τ — both curves above (free-radical conversion vs. time, step-growth extent of reaction vs. time) are exactly that PFR/batch curve. A CSTR, fed continuously and mixed instantaneously, instead runs its *entire* volume at one steady-state (outlet) composition — so for these kinetics, where the rate falls as the reactant is consumed, a single CSTR is always less efficient than a PFR/batch at the same residence time: it spends the whole time reacting at the low outlet concentration instead of starting fast at the feed concentration. Both panels below reuse the sliders from sections 5 and 6 — see `reactors.jl`.



8. Coordination polymerization kinetics

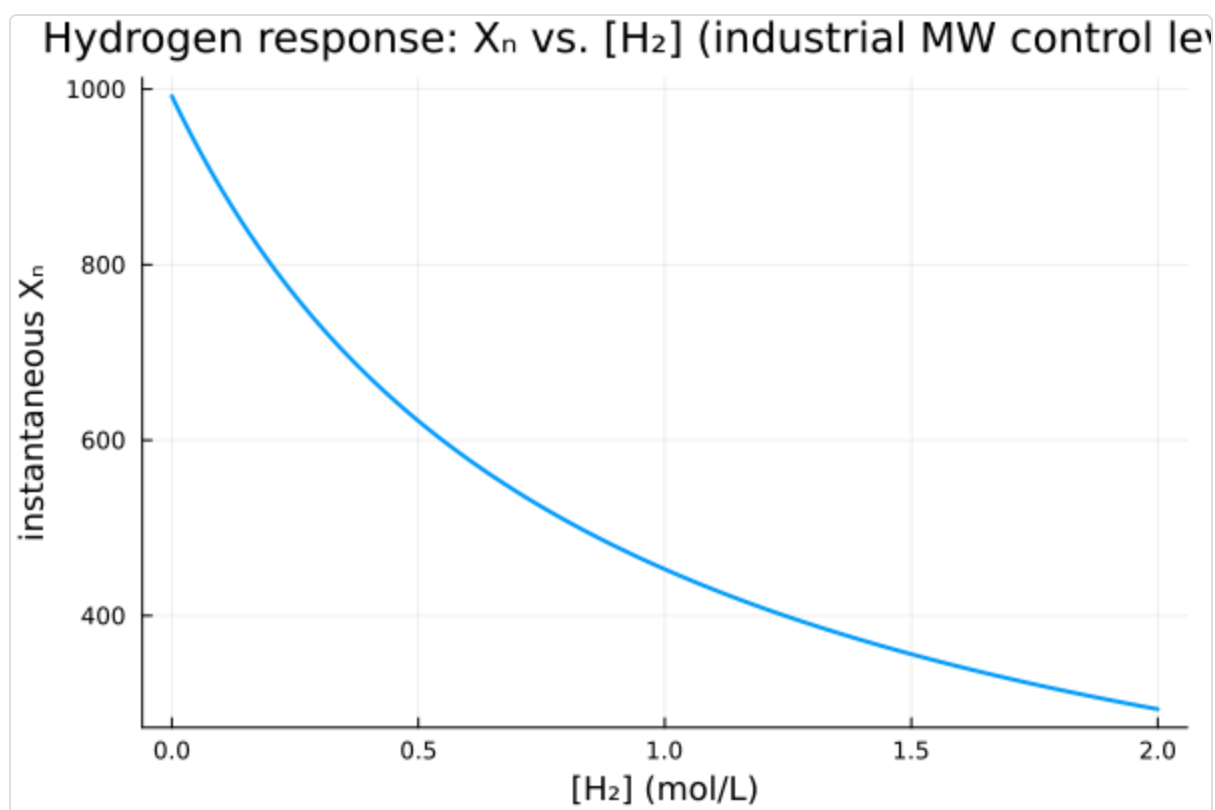
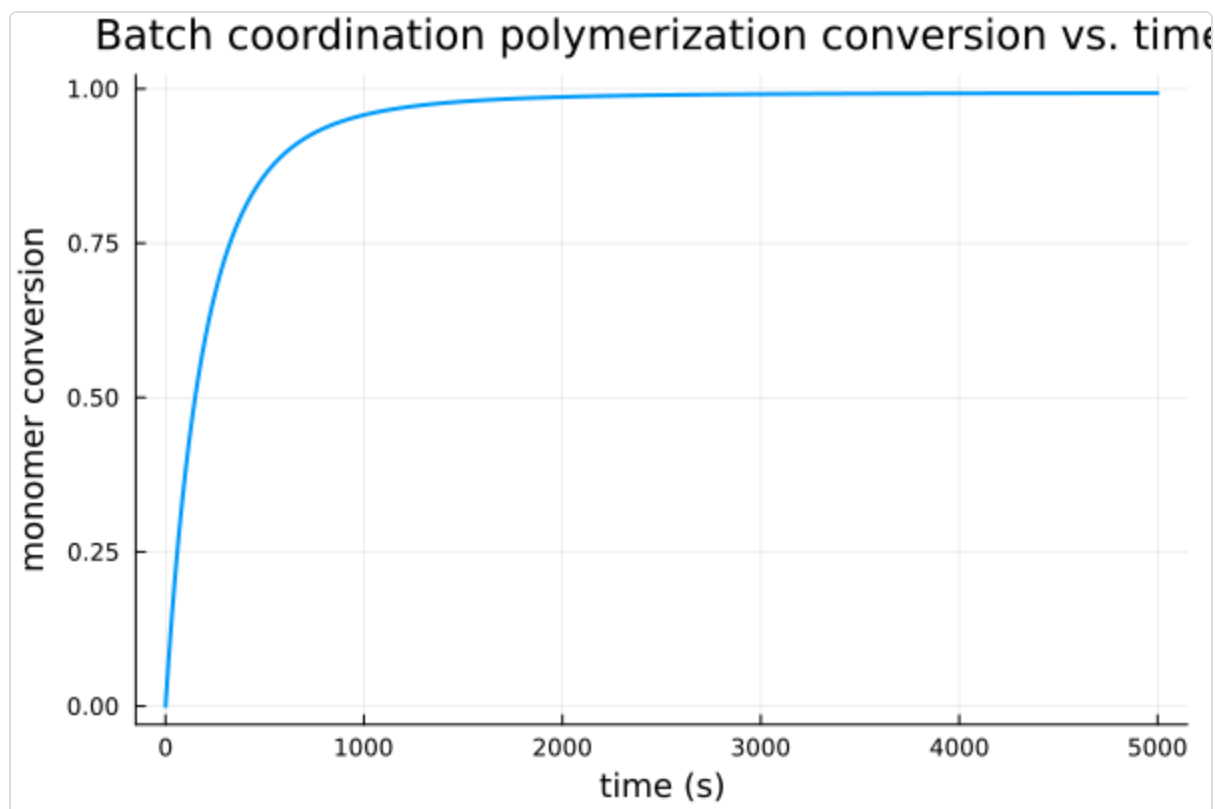
Ziegler-Natta / metallocene-type coordination polymerization, from `kinetics_coordination.jl`.

Unlike free-radical termination, chain transfer here doesn't kill the active site — it releases one dead chain and immediately starts a new one — so X_n is set by the ratio of propagation to the *total* rate of chain-releasing events (transfer + true termination), not a bimolecular termination step. Vary the active site concentration $[C]_0$, the *true-termination rate* k_t ($0 = \text{fully "living", no site deactivation}$), and the hydrogen concentration $[H_2]$ — *industrially, H_2 is dosed specifically to *lower* molecular weight via chain transfer to hydrogen.

$$[C^*]_0 = 0.0001 \text{ mol/L}, k_t = 0.001 \text{ 1/s}, [H_2] = 0.1 \text{ mol/L}$$

Instantaneous $X_n = 886.5$ (at the initial monomer concentration)

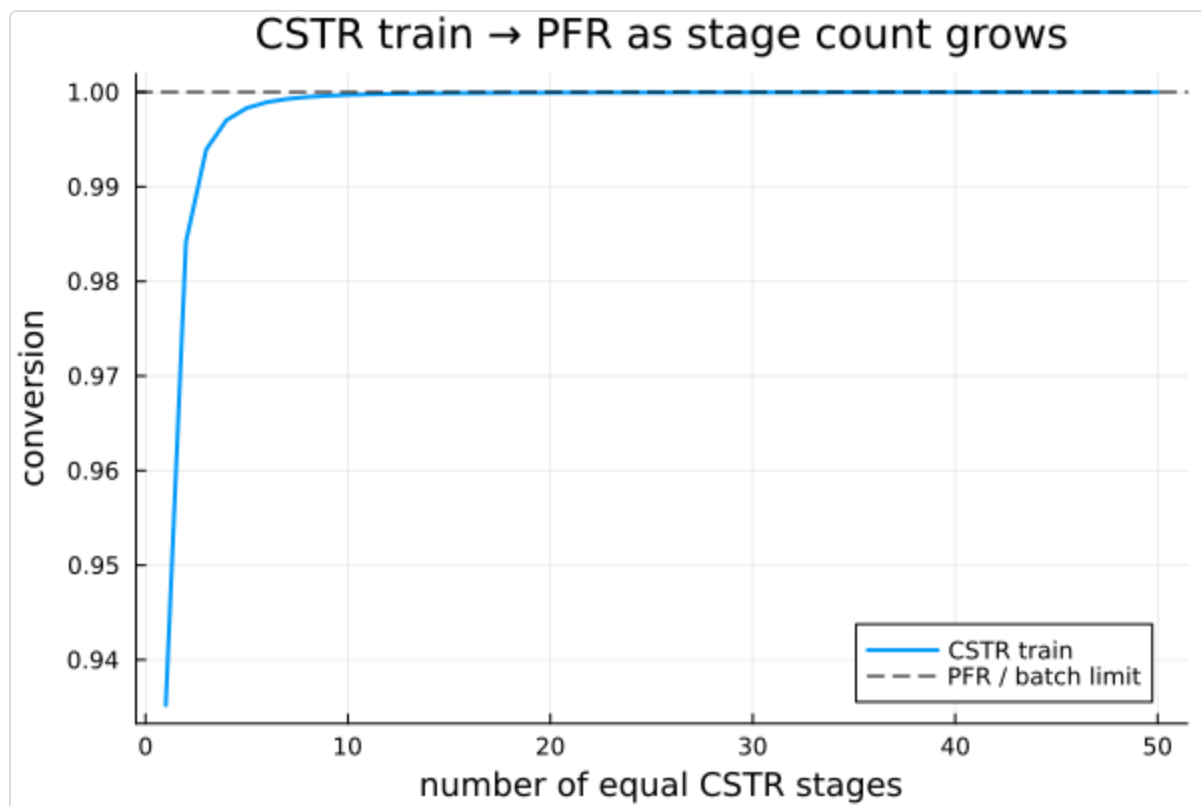
The active-site pool decays with a half-life of 693.0 s, so conversion will plateau below 100% (a "dead-end" polymerization, like the free-radical case in section 5).



9. Reactor trains: CSTRs in series → PFR

Chaining n ideal CSTRs in series, each stage's outlet feeding the next stage's inlet (`reactors.jl`), and holding the *total* residence time fixed while increasing n : performance should climb monotonically from the single-CSTR value toward the PFR/batch value as $n \rightarrow \infty$ — a classic reactor-engineering result. Reuses the free-radical parameters from section 5, with total residence time equal to that section's time axis.

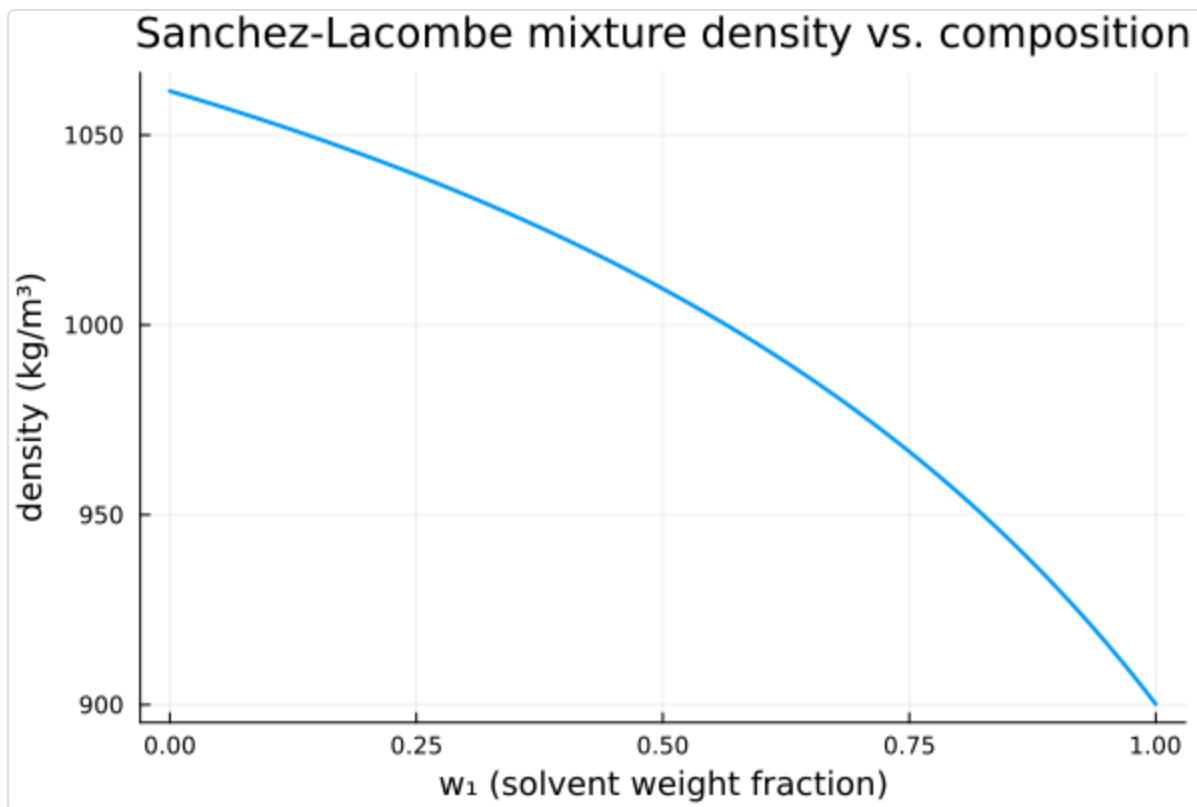
At fixed total residence time $\tau = 500000.0$ s: a single CSTR gives conversion = 0.9352; 5 equal CSTRs in series give 0.9983; the PFR/batch limit is 1.0.



10. Sanchez-Lacombe binary mixture PVT

Mixture density from the Sanchez-Lacombe mixing rules (`sl_mixing_rules` , `sl_mixture_species` , `sanchez_lacombe_mixture.jl`), applied to the same polymer/solvent pair and temperature chosen in section 1 and the pressure from section 4. w_1 is the *weight fraction of solvent* in the mixture. Only PVT (density) behavior is covered here; mixture chemical potentials/activities use a *different* (thermodynamically consistent) mixing-rule convention — see section 13.

50.0% Toluene / 50.0% Polystyrene by mass, at 25 °C, $P = 0.1$ MPa: mixture density = 1009.5 kg/m³ (pure Toluene: 900.1 kg/m³; pure Polystyrene: 1061.6 kg/m³).

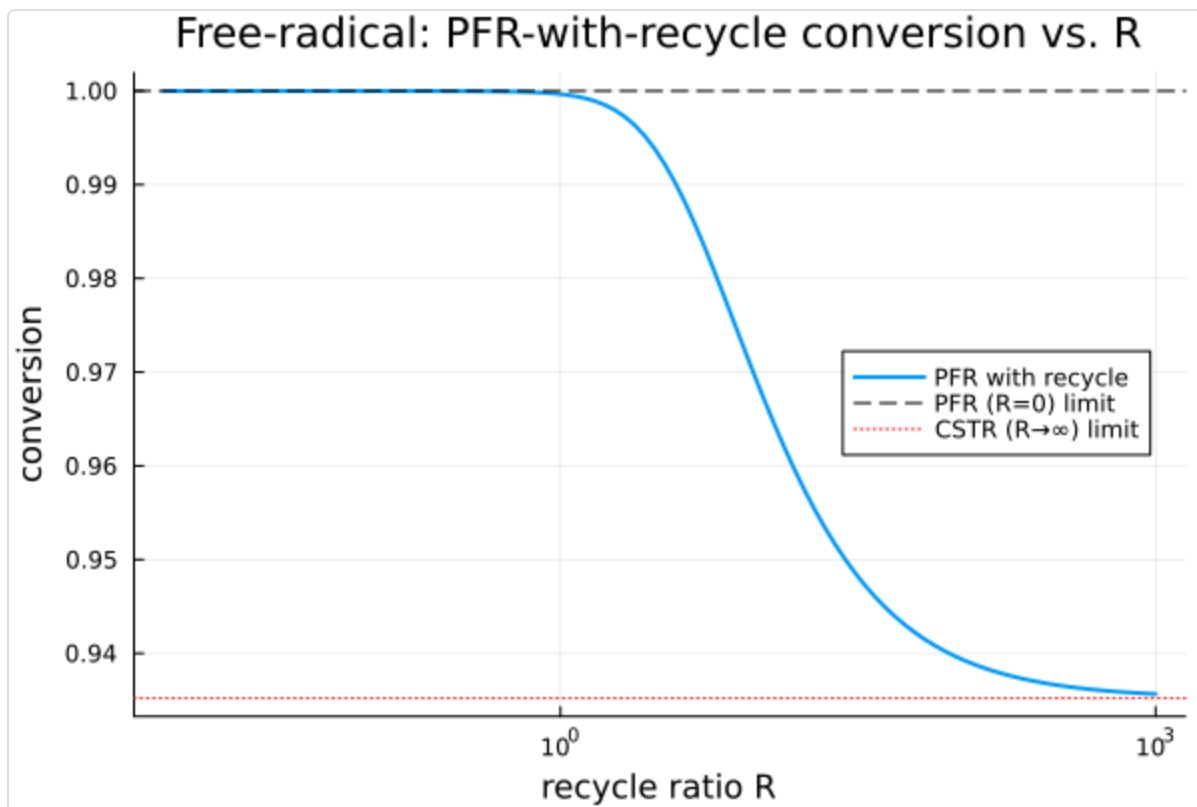


11. PFR with recycle

A single ideal PFR whose outlet stream is split, with a fraction $R = Q_{\text{recycle}}/Q_{\text{fresh}}$ sent back and mixed into the fresh feed (`reactor_recycle.jl`) — the first step beyond a single pass toward genuine flowsheet topology, since the recycle stream's composition depends on the very reactor outlet it feeds, making this an implicit ("flowsheet-style") problem. At $R = 0$ this is exactly the plain PFR curve from sections 7/9 above; as $R \rightarrow \infty$ it approaches the CSTR at the *same* nominal τ — a classic reactor-engineering result (a PFR with infinite recycle behaves like a CSTR), checked directly in the test suite. Reuses the free-radical parameters and total residence time from sections 5 and 9, the coordination parameters from section 8, and the step-growth external-catalyst parameters from section 6.

At $R = 1.0$:

- Free-radical ($\tau = 500000.0$ s): conversion = 0.9997 (PFR limit at $R=0$: 1.0; CSTR limit at $R \rightarrow \infty$: 0.9352)
- Coordination ($\tau = 100.0$ s): conversion = 0.3436 (PFR limit: 0.3786; CSTR limit: 0.3125)
- Step-growth, external catalyst ($\tau = 50$): $p = 0.9304$ (PFR limit: 0.9615; CSTR limit: 0.819)

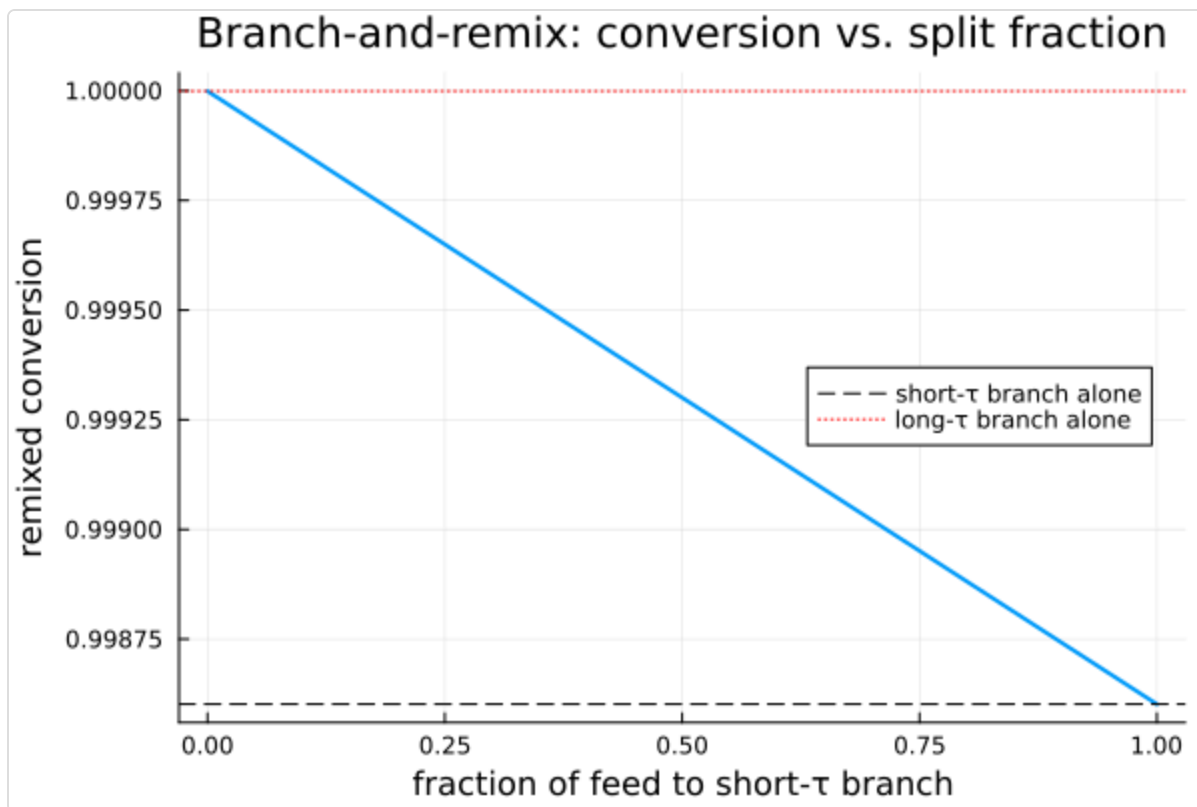


12. General flowsheet: branch-and-remix

Beyond a single recycle loop (section 11), `flowsheet.jl`'s `Stream`, `mix`, and `split_stream` primitives compose into *any* topology — this one doesn't even need `solve_tear`, since there's no cycle to converge. The free-radical fresh feed (section 5) is split into two branches, each run through its own PFR at a different residence time, then remixed — plain Julia function composition over `Streams`, reusing the same `pfr_free_radical` reactor function as everywhere else (via its `Stream`-taking method).

50.0% of the feed to the short- τ branch ($\tau = 125000.0$ s, conversion = 0.9986), remainder to the long- τ branch ($\tau = 1.0e6$ s, conversion = 1.0).

Remixed product conversion = 0.9993 — exactly the flow-weighted average of the two branch conversions (a mixer's own definition, `mix`), so it always lies between them.



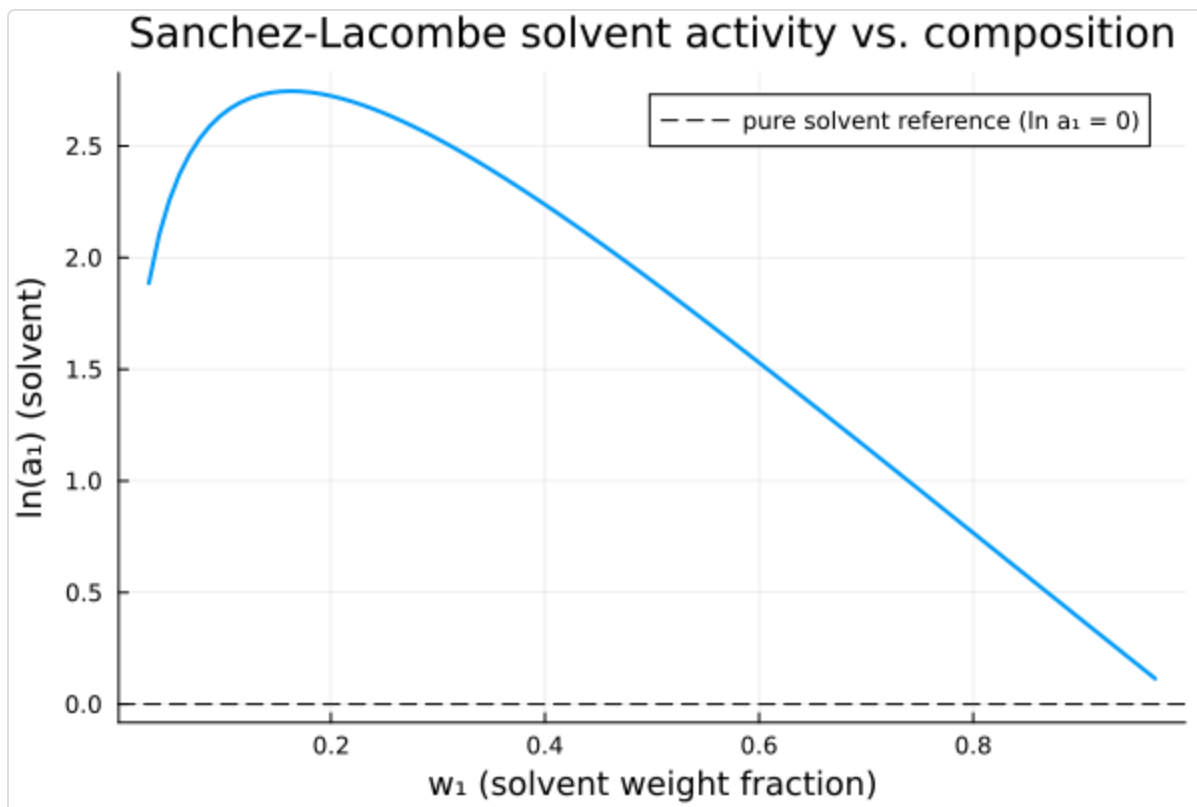
13. Sanchez-Lacombe mixture activities (consistent, constant-hole-volume)

Section 10 covered mixture *density*; this section covers mixture *chemical potentials*, via `sl_mixture_activities` (`sanchez_lacombe_activity.jl`), following von Konigslow, Park & Thompson (2017): the Sanchez-Lacombe mixture free energy gives thermodynamically consistent chemical potentials only when the hole volume is held constant with respect to composition under the derivative — implemented here with its own, separately verified mixing rules (**not** `sl_mixing_rules` from section 10 — see the module docstring for why those can't be reused). Each component's activity is reported relative to its own pure fluid at the same T, P, reusing the same polymer/solvent pair, temperature, and pressure as sections 1 and 4.

50.0% Toluene / 50.0% Polystyrene by mass, at 25 °C, P = 0.1 MPa:

Solvent (Toluene) activity: $\ln(a_1) = 1.8959$, $a_1 = 6.6586$

Polymer (Polystyrene) activity: $\ln(a_2) = -2998.0$ (per mole of chain — astronomically small for a long polymer chain, same "per mole of chain" convention as `ln_activity_polymer` in the Flory-Huggins section)



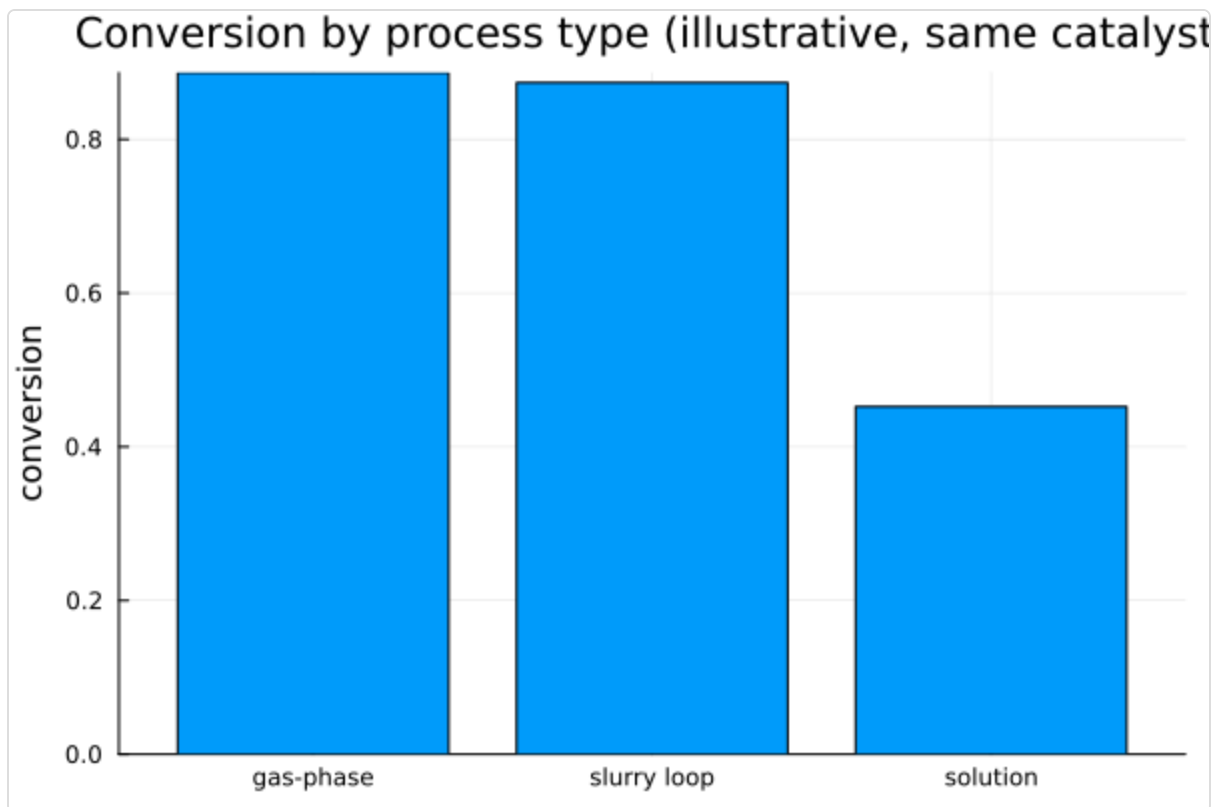
14. Industrial polyolefin processes: gas-phase, slurry-loop, solution

Polyethylene and polypropylene (`species("polyethylene")`, `species("polyethylene_ldpe")`, `species("polypropylene")` — section 1) are made industrially by several reactor configurations that need no new kinetics beyond `kinetics_coordination.jl`, just the right reactor model and, for the gas-phase case, converting a monomer *partial pressure* to the concentration the kinetics expect (`ideal_gas_concentration`). All three below share the same illustrative propagation/deactivation rate constants and active-site concentration — only the residence time and (for the gas-phase case) the monomer supply differ, matching how these processes actually differ operationally. See the README's process-mapping table for typical temperature/pressure context (not modeled explicitly here — this package's coordination kinetics take `k_p`, `k_t` as given, without a built-in Arrhenius temperature dependence).

Gas-phase: $[M] = 0.497$ mol/L from $P = 1.5$ MPa at 90.0 °C, $\tau = 120$ min $\rightarrow$ conversion = 0.8867

Slurry loop: $[M] = 3.0$ mol/L, $\tau = 75$ min $\rightarrow$ conversion = 0.8738

Solution: $[M] = 2.0$ mol/L, $\tau = 3.0$ min $\rightarrow$ conversion = 0.4523



15. LDPE high-pressure autoclave & tubular reactors

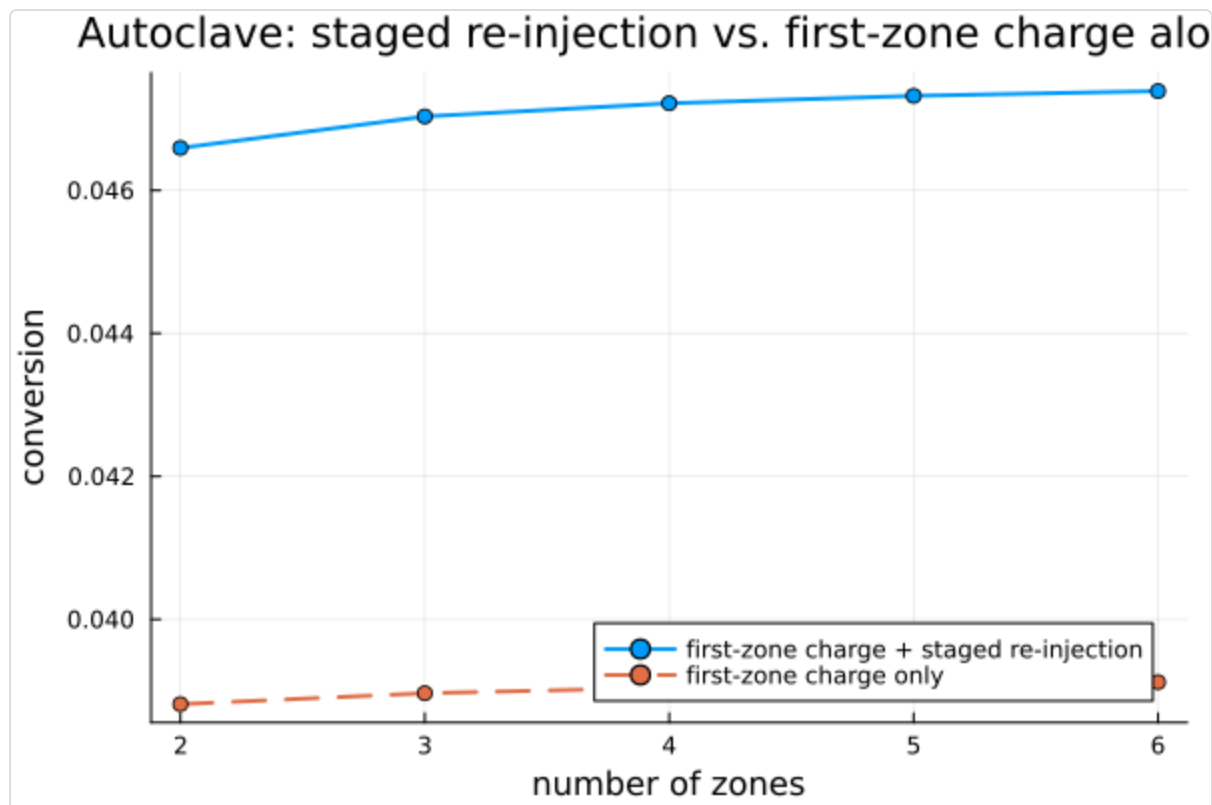
The two high-pressure (150-300 MPa) free-radical LDPE processes, both needing *staged* initiator injection to sustain the radical population along the reactor (`reactor_staged_injection.jl`): a multi-zone **autoclave** (each zone an ideal CSTR, [cstr_train_free_radical_staged](#)) and a **tubular** reactor with injection points along its length (each segment a PFR, [pfr_train_free_radical_staged](#)). `n` zones/segments share the same total residence time; `I_first` is charged to the first zone (as much as could safely be charged in a single dose — dumping the *whole* charge at once risks a runaway exotherm this package doesn't model), and an *additional* `I_extra_total` is split evenly across the remaining zones. Reuses illustrative rate constants of the same order of magnitude as section 5 (real high-pressure LDPE rate constants differ — see the note in section 14).

Note: staging the *same* total initiator differently doesn't always increase conversion (which portion reacts fastest depends on the nonlinear radical-concentration/decay coupling) — but adding *more* total initiator, staged in on top of the first dose, never decreases it. That's the comparison below, and it's also the more industrially accurate one: staging is what lets a plant add more initiator overall than a single safe dose would allow, not a free way to redistribute a fixed charge.

With 3 zones/segments, a first-zone charge of 0.01 mol/L, and 0.01 mol/L more split evenly across the rest:

Autoclave: conversion = 0.047 with staged re-injection vs. 0.039 with only the first-zone charge

Tubular: conversion = 0.0475 with staged re-injection vs. 0.0393 with only the first-zone charge



16. Case study: Spheripol-style PP (loop reactors → gas-phase impact copolymer)

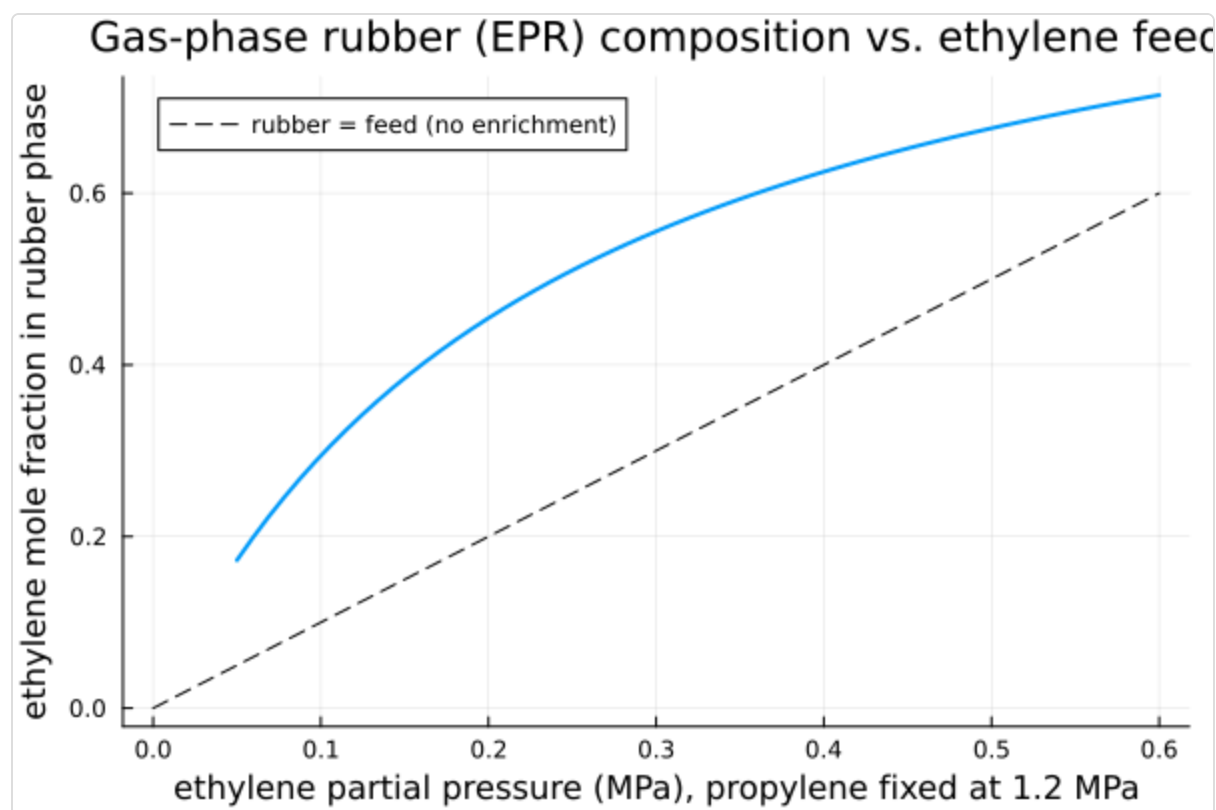
A worked example combining several pieces above into one realistic industrial flowsheet: **two liquid-propylene loop reactors in series** (the homopolymer/random-copolymer matrix — coordination catalysis, both loops sharing the same catalyst active sites via [cstr_train_coordination](#), since it's the same liquid-propylene process fluid throughout), followed by **one or two gas-phase reactors** building an ethylene-propylene rubber (EPR) phase for an *impact* copolymer grade — the catalyst's active sites (C_{star}) carry over physically from the loops into the gas phase, but the gas phase is a genuinely different reactor with its own ethylene/propylene gas feed (via [ideal_gas_concentration](#)), **not** a continuation of the loops' liquid monomer concentration — an earlier draft of this example got that wrong by chaining [cstr_train_coordination](#) straight across the phase change, caught by checking the numbers

rather than assumed. [instantaneous_copolymer_composition](#) (Mayo-Lewis) gives the resulting rubber phase's ethylene content from the gas feed composition and illustrative reactivity ratios — ethylene reacts much faster than propylene with Ziegler-Natta catalysts, so the rubber ends up markedly richer in ethylene than the gas feed, the qualitative behavior real impact-PP processes rely on.

Loops (2×60 min, propylene): conversion = 0.9069, active sites remaining = 54.1% of the fresh charge

Gas phase (1 stage(s), C_2 = partial pressure 0.3 MPa, C_3 = 1.2 MPa): conversion = 0.5887, ethylene feed fraction = 20.0%, ethylene content of the rubber phase = 55.6%

Overall product: rubber (EPR) content = 4.1% of total polymer produced



Everything originally on this package's roadmap (recycle loops, coordination reactors, a general flowsheet solver, and Sanchez-Lacombe mixture chemical potentials/activities) is now implemented above. See the repository README for details, including the trade-offs each of these made along the way.